

Analyzing Time-Varying Scalar Fields using Piecewise-Linear Morse-Cerf Theory

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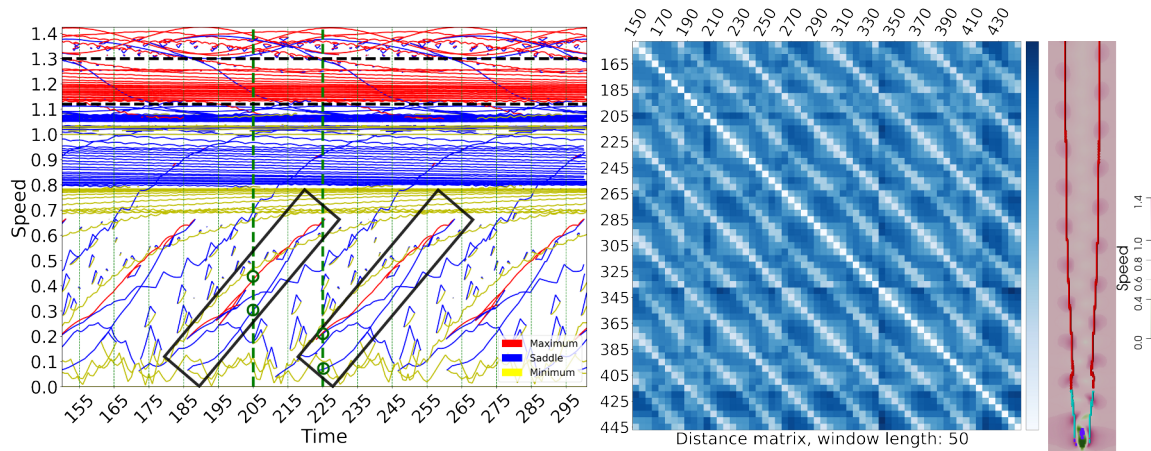


Figure 1: Analyzing periodicity and types of topological features in the von Kármán vortex street dataset using the Cerf diagram (left). Three types of maxima – sinusoidal pattern, horizontal lines in the range 1.13-1.3, and periodic in black boxes – and corresponding spatial tracks (right; red, cyan and blue). Diagonal patterns in the distance matrix (middle) reveal the periodicity.

ABSTRACT

Morse-Cerf theory considers a one-parameter family of smooth functions defined on a manifold and studies the evolution of their critical points with the parameter. This paper presents an adaptation of Morse-Cerf theory to a family of piecewise-linear (PL) functions. The vertex diagram and Cerf diagram are introduced as representations of the evolution of critical points of the PL function. The characterization of a crossing in the vertex diagram based on the homology of the lower links of vertices leads to the definition of a topological descriptor for time-varying scalar fields. An algorithm for computing the Cerf diagram and a measure for comparing two Cerf diagrams are also described together with experimental results on time-varying scalar fields.

1 INTRODUCTION

The study of time-varying scalar fields is pivotal in understanding complex dynamical systems. Topological analysis of a scalar field typically proceeds by viewing the field as a Morse function and constructs Morse-theoretic descriptors such as the Morse-Smale complex or Reeb graph [14, 26]. A sequence of elementary moves, including the cancellation or creation of a pair of critical points, can be used to transform one Morse function into another. These elementary moves applied on a topological descriptor enable the development of algorithms for simplification, comparison, and feature identification. Cerf theory is a natural extension of this approach to

one-parameter family of smooth functions [2, 1]. A key construct in this theory is the Cerf graphic of a generic one-parameter family of smooth functions, which tracks the critical values as the parameter varies. This theory however, is developed in the context of smooth functions and a PL analog is not yet available. Such a transportation of ideas to a family of PL functions is necessary in order to apply these ideas towards the analysis of time-varying scalar fields or other one-parameter families such as ensemble data.

In the context of time-varying scalar fields, previous work has described methods for computing the evolution of topological descriptors with a focus on characterizing changes to the combinatorial representation, including time-varying Reeb graph [8], merge tree [19], nested tracking graph [16], and time-varying extremum graph [4]. Comparison measures between topological descriptors [23, 22, 26, 20, 15, 25] have been used to track critical points over time. Other approaches study the evolution of persistence diagrams, including vineyards [3] and multi-parameter persistence modules arising from a one-parameter family [1]. While these descriptors support the tracking of critical points, no known descriptor provides a comprehensive representation of all topological events for PL functions together with a supporting theoretical foundation.

Contributions. In this paper, we initiate the study of time-varying scalar fields on a combinatorial manifold using a piecewise-linear (PL) adaptation of Cerf theory. Our approach involves the development of two key constructs, vertex diagram and Cerf diagram, which enable the study of the temporal behavior of critical points. We study degeneracies in the scalar field, represented as crossings in the vertex diagram, and analyze changes in the time-varying Betti number $\beta_i^t(v)$ associated with a critical point v . We also develop a simple algorithm for efficient computation of the Cerf diagram.

Next, we introduce the notion of a time-varying Euler Characteristic Curve (TV-ECC) for a time-varying scalar field. TV-ECC is a topological descriptor that is defined as an aggregate of its local

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variant (local TV-ECC), defined at each vertex. This descriptor is used to define a comparison measure between two Cerf diagrams. Finally, we present experimental results on two datasets to show the potential utility towards the analysis of time-varying scalar fields.

2 BACKGROUND

We introduce the necessary terms and concepts from Morse theory for PL functions and Cerf theory for smooth functions [7, 5]. We refer the reader to classic texts and surveys on smooth Morse theory [17] and Cerf theory [11, 13] for a detailed exposition.

2.1 PL functions on combinatorial manifolds

A simplicial complex K is called a *triangulation* of a manifold $\mathbb{M}$ if the underlying space of K is homeomorphic to $\mathbb{M}$. A PL function on $\mathbb{M}$ is defined by assigning real values to vertices of K and extending the map linearly in the interior of each simplex. A triangulated manifold of dimension d is called a *combinatorial manifold* if the link of each vertex is homeomorphic to a sphere of dimension $d - 1$.

Throughout this paper, we work in the category of combinatorial manifolds and analyze topological properties of one-parameter families of PL functions on them. First, we recall some basic notions associated with a PL function $f : \mathbb{M} \rightarrow \mathbb{R}$ [7]. The *star* of a vertex $u \in K$ is defined as $\text{st}(u) = \{\sigma \in K : u \in \sigma\}$. The *link* of a vertex u is defined as $\text{lk}(u) = \{\tau \in K : \exists \sigma \in \text{st}(u), \tau \subset \sigma, u \notin \tau\}$. The *lower link* of a vertex u is a subset of the link and is defined as $\text{lk}^-(u) = \{\tau \in \text{lk}(u) : \forall v \in \tau, f(v) \leq f(u)\}$.

The PL function f is called *generic PL* if $f(u) \neq f(v)$ for two distinct vertices u, v of K . Vertices of K may be classified as regular or critical with respect to a generic PL function f based on the homology of the lower link. They are also referred to as “homologically regular” or “H-regular” and “homologically critical” or “H-critical” [12, Definition 3.3].

Definition 2.1 (Regular point). A vertex v is called a *regular point* of f if for every $i \in [0, d]$, $\tilde{H}_{i-1}(\text{lk}^-(v); \mathbb{R}) = 0$.

Definition 2.2 (Critical point). A vertex v is *critical* for f if the homological index $\beta(v) = (\beta_0(v), \beta_1(v), \dots, \beta_d(v))$, where $\beta_i(v) = \dim_{\mathbb{R}} \tilde{H}_{i-1}(\text{lk}^-(v); \mathbb{R})$, is not identically zero.

In the above, we adopt the convention that the reduced homology group $\tilde{H}_{-1}(\mathbb{X}; \mathbb{R}) = 0$ if $\mathbb{X} \neq \emptyset$, and $\tilde{H}_{-1}(\mathbb{X}; \mathbb{R}) = \mathbb{R}$ if $\mathbb{X} = \emptyset$. A critical point v is called *non-degenerate* if exactly one entry $\beta_i(v)$ of its index $\beta(v)$ equals 1 and all others are 0. In this case, v is called a critical point of index i . If exactly one entry β_i is non-zero then the critical point of index i is said to have multiplicity β_i .

Definition 2.3 (PL Morse function). A generic PL function, whose critical points are all non-degenerate, is called a *PL Morse* function.

2.2 Cerf theory for smooth functions

A generic one-parameter family of smooth functions $f_t : \mathbb{M} \rightarrow \mathbb{R}$, where t ranges over an interval I , is Morse at all but finitely many time instances. At a degenerate time instance t_0 two critical points of f_t may have identical values. As time t increases and crosses t_0 , the corresponding pair of critical values may cross each other. Alternatively, we observe a birth/death transition of critical points. The function f_t can be expressed as follows in terms of local coordinates $x = (x_1, \dots, x_d)$ of $\mathbb{M}$ within the neighborhood of a degenerate birth/death critical point [2].

$$f_t(x) = x_{k+1}^3 \pm tx_{k+1} - \sum_{i=1}^k x_i^2 + \sum_{i=k+2}^d x_i^2. \quad (1)$$

The cubic term in x_{k+1} unfolds the degenerate quadratic so that at $t = 0$ a canceling pair of index- k and index- $(k+1)$ critical points is created or annihilated, as determined by the $\pm$ sign before x_{k+1} .

Suppose that each f_t takes values in a compact interval $J \subset \mathbb{R}$ for $t \in I$. Cerf studied the stratification of the space of smooth functions on $\mathbb{M} \times I$ [2].

Definition 2.4 (Cerf graphic). The *Cerf graphic* of a smooth family of functions $\{f_t\}$ is the subset

$$\mathcal{C} = \{(t, c) \in I \times J \mid c \text{ is a critical value of } f_t\} \subset I \times J.$$

3 PL MORSE-CERF THEORY

PL Morse theory analyzes a combinatorial manifold’s topology using PL Morse functions. We introduce an extension of Cerf theory to the PL category aimed at developing topological descriptors of PL time-varying scalar fields and methods for analyzing such fields.

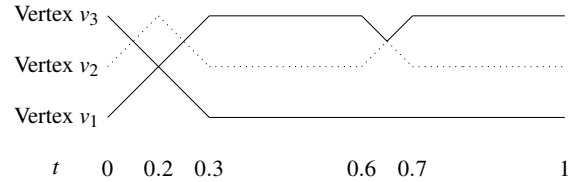


Figure 2: Vertex diagram of a PL time varying function represents the evolution of regular (dotted) and critical points (solid).

Diagrams. Let $\mathbb{M}$ be a d -dimensional combinatorial manifold and V be the vertex set of its triangulation K . We will consider the time interval $I = [0, 1]$. A one-parameter PL family $\{f_t\}$ is called generic if the collection of vertex curves $\{(t, f_t(v)) \mid t \in [0, 1]\}$ for all $v \in V$ contains finitely many degree-2 intersection points at distinct time steps $0 < T_1 < T_2 < \dots < T_n < 1$ and contains no higher degree intersection point.

Definition 3.1 (Vertex diagram). Given a generic one-parameter PL family $\{f_t\}$, the set $\{(t, f_t(v)) \mid t \in [0, 1], v \in V\}$ is called the *vertex diagram* of $\{f_t\}$.

Figure 2 shows a vertex diagram, consisting of three vertices, that distinguishes between critical and regular points. Next, we introduce the notion of a PL Morse family.

Definition 3.2 (One-parameter PL Morse family). A one-parameter PL family $\{f_t\}$ is called a *PL Morse family* if in addition to being generic, f_t is PL Morse for $t \neq T_1, \dots, T_n$.

Note that being PL Morse is not a generic property of a one-parameter PL family. However, in practice any such family can be approximated by a PL Morse family [9]. Hence, theoretical results derived for PL Morse families are useful for analyzing time-varying scalar fields.

Definition 3.3 (Cerf diagram). Let $C_t \subset V$ be the set of critical points of f_t . The *Cerf diagram* of the family $\{f_t\}$ is given by the set $\{(t, f_t(v)) \mid t \in [0, 1], v \in C_t\}$.

Figure 3 shows the Cerf diagram of a synthetic sum-of-Gaussians dataset. We denote by $\text{lk}_t^-(v)$ the lower link of the vertex v with respect to f_t , and define time-varying Betti numbers as $\beta_i^t(v) := \dim_{\mathbb{R}} \tilde{H}_{i-1}(\text{lk}_t^-(v); \mathbb{R})$.

Crossings. The vertex diagram encodes sufficient information to derive the tracking graph of critical points, and hence enables the tracking of the topological features associated with critical points. We present an exhaustive list of the types of crossings between two time instances t_1 and t_2 in a vertex diagram of a PL Morse family. In each case, we also describe the effect on the tracking graph within the time interval $[t_1, t_2]$.

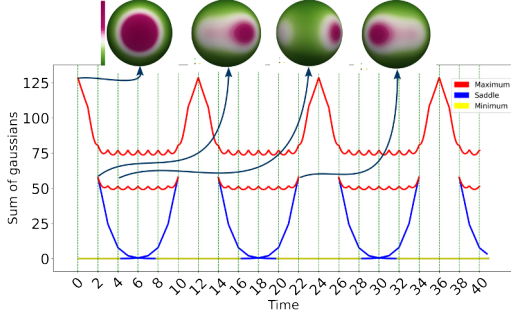


Figure 3: Cerf diagram of a synthetic sum-of-Gaussians scalar field, where the centres of the two Gaussians move in opposite directions along the equator of a sphere at a speed of $\pi/12$. We observe a (half) period of 12 from the Cerf diagram because the centres meet at the opposite pole. The centres split at $t = 0$, and a second maximum appears at $t = 2$ resulting in a birth event. Both maxima are visible at $t = 4$, the centres return to their starting position at $t = 24$, soon after the second maximum dies at $t = 22$.

Critical-critical crossing: Two critical points cross without a change in their indices; no corresponding vertex movement occurs in the tracking graph.

Regular-regular crossing: Two regular points cross and remain regular; no corresponding vertex movement in the tracking graph.

Critical-regular crossing: A critical point and a regular point cross following which neither transitions between critical and regular; no corresponding vertex movement in the tracking graph.

Critical-regular switch: A critical point and a regular point cross, then both transition; in the tracking graph, the feature at the critical point before the crossing moves linearly toward the second vertex.

Critical-critical index swap: Two critical points cross and remain critical but their Morse indices interchange; in the tracking graph, the feature at each vertex moves linearly to the other vertex.

Birth crossing: Two regular points cross and both transition to become critical; in the tracking graph, two new features are born at time of crossing at the two vertices.

Death crossing: Two critical points cross and both become regular; in the tracking graph, the two corresponding features die at the time of crossing.

The Betti number $\beta_i^f(v)$ of the lower link of a vertex v changes after a crossing. We characterize this change as follows.

Theorem 3.4. *Let $\{f_t\}$ be a generic one-parameter PL family. If two vertices v and u cross between time instances t_1 and t_2 in the vertex diagram of $\{f_t\}$ and there is no other crossing within the time range $[t_1, t_2]$, then the Betti numbers of the lower link of v and u satisfy the following relationship:*

$$\sum_{i=0}^d (-1)^i (\beta_i^{t_2}(v) - \beta_i^{t_1}(v)) = - \sum_{i=0}^d (-1)^i (\beta_i^{t_2}(u) - \beta_i^{t_1}(u))$$

Proof. Let $\text{lk}_{t_1}^-(v)$ and $\text{lk}_{t_1}^-(u)$ be the lower links of vertices v and u at time t_1 , and I_{uv} denote their intersection, $I_{uv} = \text{lk}_{t_1}^-(v) \cap \text{lk}_{t_1}^-(u)$. Without loss of generality, assume that $f_{t_1}(v) > f_{t_1}(u)$. If vertices v and u are connected by an edge of K , then at a time instance t_2 after the crossing, their lower links may be expressed as follows:

$$\text{lk}_{t_2}^-(u) = \text{lk}_{t_1}^-(u) \cup (v \star I_{uv}),$$

$$\text{lk}_{t_2}^-(v) = \text{lk}_{t_1}^-(v) \setminus (u \star I_{uv}).$$

Applying the Mayer-Vietoris sequence to the decomposition

$\text{lk}_{t_2}^-(u) = \text{lk}_{t_1}^-(u) \cup (v \star I_{uv})$ yields the long exact sequence

$$\begin{aligned} \cdots \rightarrow \tilde{H}_n(I_{uv}) \rightarrow \tilde{H}_n(\text{lk}_{t_1}^-(u)) \oplus \tilde{H}_n(v \star I_{uv}) \\ \rightarrow \tilde{H}_n(\text{lk}_{t_2}^-(u)) \rightarrow \tilde{H}_{n-1}(I_{uv}) \rightarrow \cdots \end{aligned}$$

Similarly, we obtain a long exact sequence for $\text{lk}_{t_2}^-(v) = \text{lk}_{t_1}^-(v) \setminus (u \star I_{uv})$. Since $v \star I_{uv}$ and $u \star I_{uv}$ are contractible, their reduced homology groups vanish, and the above exact sequences simplify to

$$\cdots \rightarrow \tilde{H}_n(I_{uv}) \rightarrow \tilde{H}_n(\text{lk}_{t_1}^-(u)) \rightarrow \tilde{H}_n(\text{lk}_{t_2}^-(u)) \rightarrow \tilde{H}_{n-1}(I_{uv}) \rightarrow \cdots,$$

$$\cdots \rightarrow \tilde{H}_n(I_{uv}) \rightarrow \tilde{H}_n(\text{lk}_{t_2}^-(v)) \rightarrow \tilde{H}_n(\text{lk}_{t_1}^-(v)) \rightarrow \tilde{H}_{n-1}(I_{uv}) \rightarrow \cdots.$$

Let r_i denote the rank of $\tilde{H}_{i-1}(I_{uv})$. Counting alternating sums of ranks in both exact sequences, we obtain,

$$\sum_{i=0}^d (-1)^i r_i + \sum_{i=0}^d (-1)^{i+1} \beta_i^{t_1}(v) + \sum_{i=0}^d (-1)^i \beta_i^{t_2}(v) = 0,$$

$$\sum_{i=0}^d (-1)^i r_i + \sum_{i=0}^d (-1)^{i+1} \beta_i^{t_1}(u) + \sum_{i=0}^d (-1)^i \beta_i^{t_2}(u) = 0.$$

Equating the two expressions, removing the common term involving r_i , and combining the remaining two terms, we obtain the desired relationship. $\square$

Topological descriptor. The evolution of the Betti numbers characterize crossings in the vertex diagram. We aggregate the evolution over all critical points to develop a topological descriptor of a time-varying scalar field.

Definition 3.5 (Local TV-ECC). Given a generic one-parameter PL family $\{f_t\}$ on $\mathbb{M}$ and a vertex $v \in V$, define the *local time-varying Euler characteristic curve* $\chi_v : \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}$ as

$$\chi_v(s, t) := \begin{cases} \sum_{i=0}^d (-1)^i \beta_i^t(v) & \text{if } f_t(v) \leq s, \\ 0 & \text{otherwise.} \end{cases}$$

Dlotko and Gurnari [6] introduced the Euler Characteristic Curve (ECC) as a shape invariant on a filtered simplicial complex, and demonstrated its use for topological data analysis. For a single parameter filtration K_s of the simplicial complex K , ECC is defined as $\text{ECC}(K, s) = \chi(K_s)$. The ECC can be expressed as a sum of χ_v restricted to a single time instance t (see supplementary material for the proof),

$$\text{ECC}(K_t, s) = \sum_{v \in V} \chi_v(s, t),$$

where K_t is the simplicial complex K together with the lower star filtration induced by f_t . The global version of χ_v is obtained as a sum over all vertices.

Definition 3.6 (TV-ECC). Given a generic one-parameter PL family $\{f_t\}$, the *time-varying Euler Characteristic Curve* $\mathcal{E}_{\{f_t\}} : \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}$ is defined as

$$\mathcal{E}_{\{f_t\}}(s, t) := \sum_{v \in V} \chi_v(s, t).$$

4 COMPUTING AND COMPARING CERF DIAGRAMS

In this section, we describe an algorithm to compute the Cerf diagram and to compare two diagrams. The input consists of the scalar field specified at each vertex v for all T time steps. The scalar values at a vertex are linearly interpolated between every pair of consecutive time steps resulting in a time-varying PL scalar field. Since the criticality of a vertex is determined by its lower link, the algorithm tracks all crossings in the vertex diagram by examining pairs of vertices that lie in the link of each other. While processing a crossing of vertices u and v , the algorithm updates $\text{lk}^-(u)$

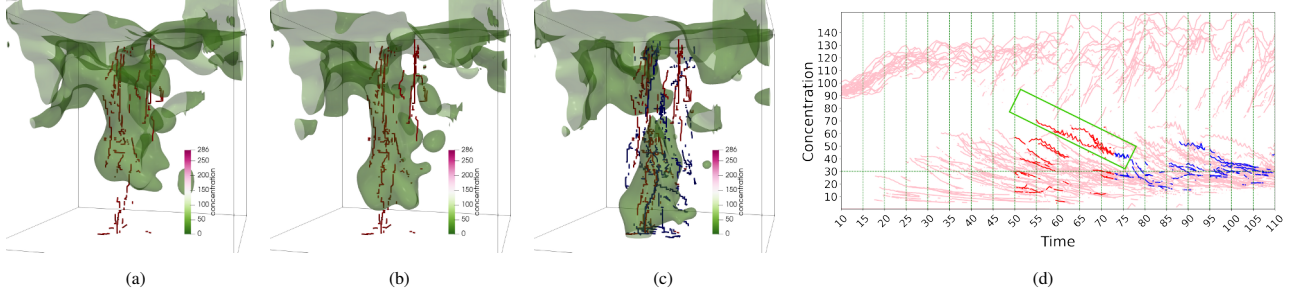


Figure 4: Studying the evolution of the central finger in the viscous fingers dataset using the Cerf diagram. Tracks of maxima extracted by a search seeded inside the finger at time (a) $t = 69$, (b) $t = 71$, and (c) $t = 73$. (d) Cerf diagram contains two groups of maxima (upper / lower). Cerf arcs corresponding to maxima tracks are highlighted, distinguishing between maxima born prior to $t = 73$ (red) and after $t = 73$ (blue).

and $\text{lk}^-(v)$ and recomputes $\beta(u)$ and $\beta(v)$ to determine the criticality of the vertices. If the vertices are critical, the corresponding arc between the vertices in the vertex diagram is declared as a Cerf arc. The Cerf diagram is represented as a collection of 6-tuples $\{(t_1, f_{t_1}(v), t_2, f_{t_2}(v), v, \beta(v))\}$ where $(t_1, f_{t_1}(v))$ and $(t_2, f_{t_2}(v))$ are the end points of a Cerf arc, v represents the vertex location of the critical point in the time interval $[t_1, t_2]$, and $\beta(v)$ is its (fixed) homological index in that time interval. The detailed algorithm is presented in the supplementary material.

We propose a distance measure between two generic one-parameter PL families and use it to compare two time-varying scalar fields. The distance between two families $\{f_t\}$ and $\{g_t\}$, $t \in I$, is defined as the aggregated difference between $\mathcal{E}_{f_t}$ and $\mathcal{E}_{g_t}$,

$$\Delta(\{f_t\}, \{g_t\}) := \int_I \int_{\mathbb{R}} |\mathcal{E}_{f_t}(s, t) - \mathcal{E}_{g_t}(s, t)| ds dt. \quad (2)$$

5 EXPERIMENTAL RESULTS

We perform computational experiments on two datasets that constitute time-varying scalar fields. The results indicate the potential utility of the Cerf diagram to identify interesting patterns of critical points. The Cerf diagram used in conjunction with the corresponding spatial tracks of selected critical points can help identify evolution of interesting features in the spatial domain. Figure 3 presents an illustration on a synthetic dataset, where the Cerf diagram provides a useful static overview of the time-varying field, captures the periodicity, and helps locate interesting time instances.

5.1 2D vortex street

The 2D von Kármán vortex street dataset is a simulation of a flow around a cylinder that exhibits periodic vortex shedding. The speed (velocity magnitude) is available as a scalar field over a 400×50 grid across 1001 time steps [24]. Figure 1 shows the Cerf diagram computed over a time window $[150, 300]$ and tracks of maxima in the spatial domain from $t = 0$ until $t = 400$. It also shows the matrix of pairwise distances (Eq. (2)) between Cerf diagrams computed for 50-step time windows starting at $t = 150$ with a shift of 5.

Identifying and classifying topological features. We focus on the maxima because they correspond to vortices, the primary feature of interest in this dataset. The Cerf diagram helps identify three types of maxima. The maxima with speed greater than 1.3 (black dashed line) that exhibit a sinusoidal pattern are highlighted in cyan in the spatial domain. They are located in the vicinity of the cylinder. The maxima with speed between 1.12 and 1.3 do not exhibit significant variation in the speed and lie further away from the cylinder. They are shown in red in the spatial domain and correspond to the vortex street. Other interesting maxima with speeds lower than 1.12 exhibit periodic birth-death behavior (black boxes) with a lifetime of ~ 37 and occur near the cylinder.

Investigating periodicity and temporal events. While the periodic nature of evolution of critical points is visible from the Cerf diagram (region with speed < 0.7), the repeating pattern is clear in the distance matrix. Both the full period (75) and half period (~ 37) can be deduced from the diagonal patterns in the matrix, and they match with previously reported results [18, 23]. The saddle-maximum pair (green circles) appear at significantly different speeds for $t = 205$ and $t = 225$, and for other pairs of time steps in the time window that starts at 205 and 225. This may be a reason why the distance matrix exhibits dark bands in the region corresponding to this pair of windows, reflecting a difference in the evolution of features within the two windows.

5.2 Viscous fingers

The viscous fingers dataset is the result of a stochastic simulation of the mixing of high concentration salt into water [21]. We analyze the 33^{rd} member of an ensemble at a smoothing length of 20 and resampled over a $101 \times 101 \times 101$ grid across 120 time steps [10]. Figure 4 shows the Cerf diagram, displaying only maxima.

Identifying and classifying topological features. The upper band of Cerf arcs, with salt concentration above 70, correspond to maxima that form near the mixing interface close to the upper boundary of the grid. The lower band of Cerf arcs correspond to maxima located in the interior of the grid that contribute to the formation of the fingers.

Investigating fingers of interest. The central finger, represented by the isosurface at value 30 in Fig. 4, elongates and eventually splits at $t = 73$. Tracks in the spatial domain (red) show the evolution of some maxima that contribute to the formation of this finger until $t = 73$. Tracks of all maxima born after time 73 are shown in blue. These tracks are extracted by a search in the vicinity of a seed point inside a finger. The corresponding arcs are highlighted in the Cerf diagram. The shorter red tracks born between $t = 50$ and $t = 55$ correspond to the formation of this finger. A key maximum contributing to this finger shows up as a prominent track (green box). This maximum is born at $t = 55$ and dies around $t = 77$ when the associated finger also disappears.

6 CONCLUSIONS

We introduced a PL adaptation of Morse-Cerf theory, including a structural characterization and methods for computing and comparing Cerf diagrams. Demonstrating the practical utility to the analysis of time-varying scalar fields requires future efforts towards the development of methods for topological simplification, the study of Cerf moves that enable local modifications of the Cerf diagram, a comprehensive analysis of runtime performance of the algorithm, and comparisons with alternative methods.

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